

Bachelor's Thesis Proposal

Formalizable Vietnam's Personal Data Protection Law

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1 Motivation

Since the beginning of 2026, organizations that process personal data in Vietnam must comply with the Law on Personal Data Protection (Law No. 91/2025/QH15) [4] and its implementing Decree No. 356/2025/ND-CP [3]. Like the GDPR [2], the law imposes strict requirements on how (personal) data is handled, but it is written in natural language, which is often ambiguous and open to interpretation.

Some of these requirements might be precise enough to be stated in formal logic. For example, "personal data may only be processed while the data subject's consent is in force" or "a data breach must be reported to the authority within a fixed deadline". For the GDPR, it has been shown that a significant part of the regulation can be stated formally [1]. For the Vietnamese law, no such analysis exists yet.

2 Objective

The objective of this thesis is to analyze a selected part of the Law on Personal Data Protection and Decree No. 356/2025/ND-CP, e.g., the provisions on consent, data subject rights, and breach notification, and to determine for each requirement whether it can be formalized.

3 Tasks

1. Read and select the subset of provisions of the law and the decree, and extract the individual requirements they impose on organizations that process personal data.
2. Review different logic systems, e.g., propositional logic, first-order logic, and temporal logic.
3. Study how the GDPR has been formalized.
4. For each requirement, decide whether it can be formalized. If not, explain why.
5. For each formalizable requirement, choose the simplest suitable logic, write the formula, and explain it in plain English.
6. Document the methodology and the requirements catalog in the final report.

4 Deliverables

The following deliverables are due at the end of the project:

Final report Please have a look at the submission requirements in the following section. The final report should consist of an introduction, a (theoretical) background section, sections describing the work packages and their outcomes, and a conclusion. The report must be written in English.

Requirements catalog The catalog of extracted requirements, each with its formalization (if any), the logic used, and a plain-English explanation, in a structured and reusable format.

Presentation At the end of the thesis, a 20-minute presentation must be given in a thesis defense seminar. It should give an overview and discuss the most important highlights of the work.

References

- [1] Emma Arfelt, David A. Basin, and Søren Debois. Monitoring the GDPR. In *Computer Security — ESORICS 2019*, volume 11735 of *Lecture Notes in Computer Science*, pages 681–699. Springer, 2019.
- [2] European Parliament and Council. Regulation (EU) 2016/679, 2016. <https://eur-lex.europa.eu/eli/reg/2016/679/oj>.
- [3] Government of the Socialist Republic of Vietnam. Decree no. 356/2025/ND-CP detailing a number of articles of the law on personal data protection, 2025.
- [4] National Assembly of the Socialist Republic of Vietnam. Law on personal data protection, law no. 91/2025/QH15, 2025. <https://english.luatvietnam.vn/dan-su/law-on-personal-data-protection-law-no-91-2025-qh15-405135-d1.html>.